6. Python SQLite Game Shop Tutorial Delete

Contents

6. Python SQLite Game Shop Tutorial Delete 1
SQL Tutorials 1
SQLite with Python Tutorials 1
Explanation: Deleting Data 1
Do: Delete Data with DB Browser for SQLite 2
MVC (Model View Controller) 2
Do: sqlite_tutorial_6_delete.py 3
Assignment Submission 7

- Follow all directions carefully and accurately.
- Think of the directions as minimum requirements.

SQL Tutorials

- https://www.w3schools.com/sql/sql_delete.asp

SQLite with Python Tutorials

- SQLite Databases with Python - Full Course – FreeCodeCamp.org
- https://www.sqlitetutorial.net

Explanation: Deleting Data

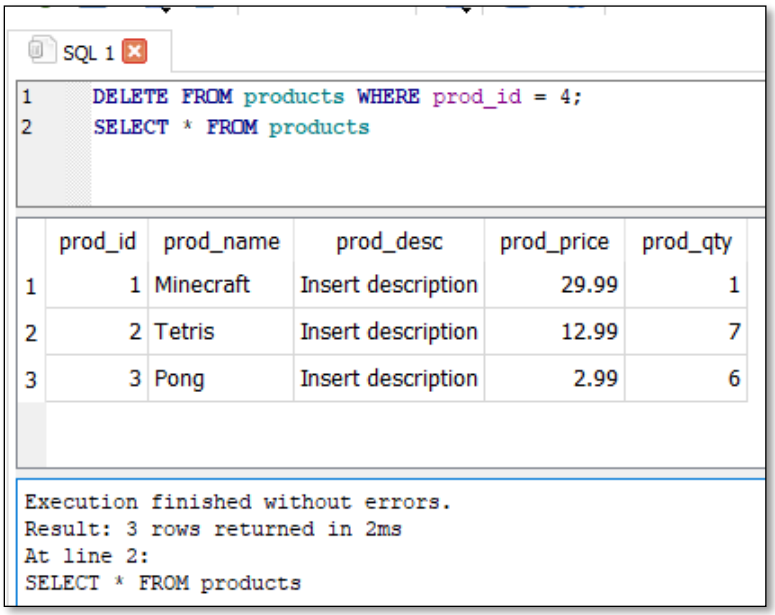
Let's say that Sammy the shark escaped into the wild. Since Sammy no longer lives at the aquarium, it would make sense to remove the Sammy row from the fish table.

```
DELETE FROM table_name [WHERE Clause]
```

```
released_fish_name = "Sammy"
cursor.execute(
    "DELETE FROM fish WHERE name = ?",
    (released_fish_name,)
)
connection.commit()
```

Do: Delete Data with DB Browser for SQLite

1. Copy **game_shop_5.db** to **game_shop_6.db**
2. In **DB Browser for SQLite** → Open Database **game_shop_6.db**
3. Go to the **Execute SQL** tab.
4. Insert and execute the following SQL code.



MVC (Model View Controller)

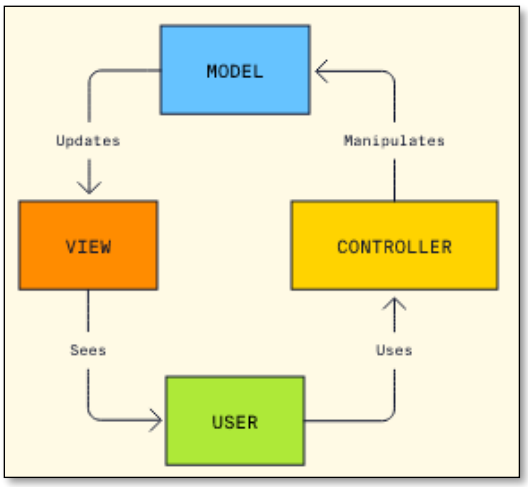
MVC is short for Model, View, and Controller. MVC is a popular way of organizing your code. The big idea behind MVC is that each section of your code has a purpose, and those purposes are different. Some of your code holds the data of your app, some of your code makes your app look nice, and some of your code controls how your app functions.

Model: Model code typically reflects real-world things. This code can hold raw data, or it will define the essential components of your app. For instance, if you were building a To-do app,

the model code would define what a "task" is and what a "list" is – since those are the main components of a todo app.

View: View code is made up of all the functions that directly interact with the user. This is the code that makes your app look nice, and otherwise defines how your user sees and interacts with it.

Controller: Controller code acts as a liaison between the Model and the View, receiving user input and deciding what to do with it. It's the brains of the application, and ties together the model and the view.



Do: sqlite_tutorial_6_delete.py

We are going to move all the database specific functions into a separate file using the MVC paradigm.

Model: game_shop_6.db

View: sqlite_tutorial_6_delete.py

Controller: db_operations.py

1. Create a Python file named: **db_operations.py**
2. Excluding the main function → Copy the rest of the methods from **sqlite_tutorial_5_insert.py** into the **db_operations.py** file.
3. Copy the main function from sqlite_tutorial_5_insert.py into **sqlite_tutorial_6_delete.py**
4. Modify **sqlite_tutorial_6_delete.py** as shown below.

```python
"""
    Name: sqlite_tutorial_6_delete.py
    Author: William Loring
    Created: 01/05/22
"""
# Import db_operations module
import db_operations


def main():
    # Create a GameShop instance with the specified database file
    game_shop = db_operations.GameShop("game_shop_6.db")

    # Create the 'products' table in the database
    game_shop.create_table()
    print("Table created")

    # Insert predefined records into the 'products' table
    game_shop.insert_records()
    print("Records inserted")

    # Fetch and display records from the 'products' table
    records = game_shop.fetch_all_records()
    print("\nRecords returned as a list of tuples")

    # Print the entire list of records
    print(records)

    # Print a specific record from the list (index 3)
    print(records[3])

    # Iterate through the list and display records one at a time
    print("Display records one at a time, iterating through list")

    # Loop through returned list of tuples
    for record in records:
        print(record)

    # Fetch all records based on the product price
    print("Fetch filtered records.")
    record = game_shop.fetch_filtered_records(.99)
    print("\nFetch all product that are $.99: ")
    print(record)

    # Update a record's price in the 'products' table based on product ID
    game_shop.update_record(prod_price="29.99", prod_id=1)
    print("\nUpdated sale price for MineCraft")
```

```python
    # fetch and display all records, showing the updated sale price
    records = game_shop.fetch_all_records()
    for record in records:
        print(record)

    # Delete a record based on the product id
    game_shop.delete_record(prod_id="4")
    print("\nRecord deleted")

    # fetch and display all records, showing the updated sale price
    records = game_shop.fetch_all_records()
    # Loop through returned list of tuples
    for record in records:
        print(record)
    print("Guessing Game is deleted")


# If a standalone program, call the main function
# Else, use as a module
if __name__ == "__main__":
    main()
```

Add the **delete_record()** method to **db_operations.py**.

```python
#------------------------- DELETE RECORD ---------------------------#
    def delete_record(self, prod_id):
        """Delete selected record."""
        with sqlite3.connect(GameShop.DATABASE) as connection:
            cursor = connection.cursor()
            SQL = "DELETE FROM products WHERE prod_id = ?"
            cursor.execute(
                SQL,
                (prod_id,)
            )
```

Example run:

```python
# ------------------------ DELETE RECORD ----------------------------------#
    def delete_record(self, prod_id):
        """
        Delete a specific product record from the 'products' table.

        Parameters:
            prod_id (int): The ID of the product to be deleted.
        """

        # Establish a connection to the database file using with
        with sqlite3.connect(self.database) as connection:

            # Create a cursor object to interact with the database
            cursor = connection.cursor()

            # SQL SELECT statement with orod_id argument substituted for ?
            SQL = "DELETE FROM products WHERE prod_id = ?"

            # Execute SQL DELETE statement with provided product ID
            cursor.execute(
                SQL,
                (
                    prod_id,
                )
            )
```

```
Table created
Records inserted

Records returned as a list of tuples
[(1, 'Minecraft', 'Insert description', 15.99, 1, 1), (2, 'Tetris', 'Insert description', 12.99, 3, 7), (
3, 'Pong', 'Insert description', 0.99, 4, 6), (4, 'Guessing Game', 'Insert description', 0.99, 10, 200)]
(4, 'Guessing Game', 'Insert description', 0.99, 10, 200)
Display records one at a time, iterating through list
(1, 'Minecraft', 'Insert description', 15.99, 1, 1)
(2, 'Tetris', 'Insert description', 12.99, 3, 7)
(3, 'Pong', 'Insert description', 0.99, 4, 6)
(4, 'Guessing Game', 'Insert description', 0.99, 10, 200)
Fetch filtered records.

Fetch all product that are $.99:
[(3, 'Pong', 'Insert description', 0.99, 4, 6), (4, 'Guessing Game', 'Insert description', 0.99, 10, 200)
]

Updated sale price for MineCraft
(1, 'Minecraft', 'Insert description', 29.99, 1, 1)
(2, 'Tetris', 'Insert description', 12.99, 3, 7)
(3, 'Pong', 'Insert description', 0.99, 4, 6)
(4, 'Guessing Game', 'Insert description', 0.99, 10, 200)

Record deleted
(1, 'Minecraft', 'Insert description', 29.99, 1, 1)
(2, 'Tetris', 'Insert description', 12.99, 3, 7)
(3, 'Pong', 'Insert description', 0.99, 4, 6)
Guessing Game is deleted
```

Assignment Submission

1. Attach the program files.
2. Attach screenshots showing the successful operation of the program.
3. Submit in Blackboard.